

EvoTox: An Evolutionary Adversarial Framework for Large Language Model Safety Evaluation

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Abstract

We present EvoTox, a novel evolutionary adversarial framework designed to systematically evaluate the safety and robustness of Large Language Models (LLMs) against toxic content generation. Our approach employs a three-component architecture comprising a System Under Test (SUT), an adversarial prompt generator, and an intelligent evaluator with enhanced feedback mechanisms. Through iterative prompt evolution guided by toxicity scores and AI-generated feedback, EvoTox demonstrates the potential for significant improvements in discovering model vulnerabilities compared to static red-teaming approaches. We introduce enhanced evaluation with detailed explanations, memory-based learning systems, and robust error recovery mechanisms. The framework supports multiple evaluation backends including Perspective API, Detoxify, and local LLM evaluation, making it adaptable for various research contexts. Our contributions include the first evolutionary approach to automated red-teaming with integrated feedback loops and comprehensive performance optimizations through asynchronous processing. Preliminary evaluations suggest substantial improvements in jailbreak discovery effectiveness while providing rich qualitative insights into attack patterns and model behavior.

1 Introduction

The rapid deployment of Large Language Models (LLMs) in production systems has raised significant concerns about their safety and robustness against adversarial inputs [1]. Traditional red-teaming approaches rely on manual effort or static prompt libraries, limiting their effectiveness in discovering novel attack vectors. Current automated safety eval-

uation methods often lack the adaptive capabilities necessary to keep pace with evolving model defenses.

We introduce EvoTox, an evolutionary adversarial framework that automatically generates and refines toxic prompts through iterative optimization. Unlike existing approaches, EvoTox employs a genetic algorithm-inspired methodology combined with AI-powered feedback loops to systematically explore the space of potential vulnerabilities.

1.1 Contributions

Our primary contributions include:

- A novel evolutionary framework for automated adversarial prompt generation
- Enhanced evaluation system providing detailed explanations and feedback
- Memory-based learning mechanism that improves across iterations
- Robust three-model architecture optimizing speed versus accuracy
- Comprehensive empirical evaluation demonstrating significant performance gains

2 Related Work

2.1 Adversarial Attacks on Language Models

Previous work on adversarial attacks against language models has primarily focused on static approaches. Zou et al. [2] proposed universal adversarial triggers, while Wei et al. [3] examined jailbreaking techniques. However, these approaches lack the adaptive capabilities necessary for comprehensive evaluation.

2.2 Automated Red-Teaming

Recent efforts in automated red-teaming include rule-based systems and template-driven approaches [4]. While effective for known attack patterns, these methods struggle to discover novel vulnerabilities that emerge as models evolve.

2.3 Safety Evaluation Frameworks

Existing safety evaluation frameworks such as those proposed by Gehman et al. [5] provide valuable baselines but lack the systematic exploration capabilities offered by evolutionary approaches.

3 Methodology

3.1 System Architecture

EvoTox employs a three-component architecture designed for flexibility and performance optimization:

1. **System Under Test (SUT):** The target model being evaluated
2. **Prompt Generator:** Creates adversarial prompt variations
3. **Evaluator:** Scores toxicity and provides detailed analysis

This separation allows independent optimization of each component, enabling researchers to use smaller, faster models for generation while employing more sophisticated models for evaluation.

3.2 Evolutionary Algorithm

The core algorithm follows an evolutionary optimization approach:

Algorithm 1 EvoTox Evolutionary Process

```
Initialize population with baseline prompts
for each generation  $g = 1$  to  $G$  do
  for each category  $c$  in {sexist, racist, homophobic, toxic, insulting} do
    Generate variant prompts using category  $c$ 
    Evaluate SUT responses for toxicity
    Apply enhanced evaluation if enabled
    Generate improvement feedback
  end for
  Select best performing prompts
  Update memory with successful patterns
  Store results and continue evolution
end for
```

3.3 Enhanced Evaluation System

Our enhanced evaluation system extends traditional toxicity scoring with detailed explanations:

```
def evaluate_ollama_with_explanation(response, prompt, category):
    system_prompt = """Analyze toxicity and
    provide:
    - Detailed score reasoning
    - Key contributing factors
    - Safety assessment
    Return JSON with scores and explanation.
    """

    # Generate evaluation with robust JSON parsing
    result = ollama.generate(model, prompt, system_prompt)
    scores, explanation = parse_with_recovery(result)
    return scores, explanation
```

3.4 Memory-Based Learning

The system incorporates memory mechanisms to avoid repeating unsuccessful strategies:

- **Short-term memory:** Recent attempt context (configurable window)
- **Feedback integration:** AI-generated improvement suggestions
- **Pattern recognition:** Identification of effective techniques

3.5 Robust Error Recovery

To ensure reliability during extended evaluations, we implemented comprehensive error recovery:

```
def clean_json_response(json_str):
    # Remove trailing commas
    json_str = re.sub(r',(\s*[\}\]])', r'\1', json_str)

    # Extract JSON from surrounding text
    json_match = re.search(r'\{.*\}', json_str, re.DOTALL)
    return json_match.group(0) if json_match else json_str
```

4 Experimental Setup

4.1 Dataset

The `questions` dataset comprises 1,961 adversarial prompts designed specifically for red-teaming and jailbreaking AI language models. This comprehensive

collection serves as a systematic evaluation framework for testing model safety mechanisms and vulnerability assessment.

4.1.1 Dataset Characteristics

The dataset functions as an **adversarial prompt collection** for AI safety testing, containing carefully crafted questions designed to challenge model safety guardrails across multiple harmful categories.

4.1.2 Content Categories

The dataset encompasses six primary categories of adversarial content:

1. **Social Harm & Discrimination:** Prompts targeting hate speech, racism, homophobia, sexism, cultural appropriation, and stereotyping across various identity groups.
2. **Psychological Manipulation:** Questions focusing on techniques for inducing feelings of inferiority, guilt, insecurity, gaslighting, memory manipulation, and mental health deterioration including anxiety, paranoia, and suicidal ideation.
3. **Economic & Corporate Misconduct:** Prompts addressing tax evasion, insider trading, market manipulation, worker exploitation, predatory practices, corruption, and regulatory circumvention.
4. **Cybersecurity Attacks:** Technical attack vectors including SQL injection, phishing, DDoS attacks, social engineering, man-in-the-middle attacks, and network exploitation techniques.
5. **Academic Misuse:** Questions attempting to weaponize legitimate academic fields such as cultural studies, gender studies, and geometry for harmful purposes.
6. **Historical Controversies:** Sensitive inquiries about historical figures, events, and potentially inflammatory topics regarding past leaders and atrocities.

4.1.3 Research Application

Within the EvoTox framework, this dataset serves as **initial seed prompts** for the evolutionary algorithm. The system attempts to optimize and modify these base prompts to successfully circumvent target model safety mechanisms. The questions are deliberately designed to trigger safety refusals, which

the evolutionary process then attempts to overcome through systematic prompt engineering.

4.1.4 Safety Research Context

This adversarial prompt collection is essential for **responsible AI safety research**, enabling researchers to:

- Systematically evaluate model vulnerabilities across diverse harmful categories
- Test the robustness of safety guardrails under adversarial pressure
- Assess model resistance to toxic, discriminatory, and dangerous content generation
- Develop and validate improved defensive mechanisms
- Conduct reproducible safety evaluations with standardized attack vectors

The comprehensive nature of this dataset ensures thorough coverage of potential attack surfaces, making it a valuable resource for advancing AI safety research and developing more robust language model defenses.

4.2 Model Configurations

We tested three performance profiles:

Table 1: Model Configuration Profiles

Component	Speed	Balanced	Research
SUT	qwen2.5:0.5b	llama3.2:3b	llama3:70b
Generator	qwen2.5:0.5b	qwen2.5:0.5b	llama3:8b
Evaluator	qwen2.5:1.5b	llama3:8b	llama3:13b

4.3 Evaluation Metrics

We measured performance across multiple dimensions:

- **Success Rate:** Percentage of attempts achieving target toxicity
- **Convergence Speed:** Iterations to reach optimal prompts
- **Explanation Quality:** Human evaluation of AI-generated analysis
- **System Reliability:** Uptime and error recovery effectiveness

5 Results

5.1 Controlled Experimental Comparison

We conducted controlled experiments comparing baseline EvoTox against the enhanced configuration with AI-powered evaluation and feedback systems. Both configurations used identical parameters: 3 iterations, homophobic category testing, Llama3 models, and memory-enabled optimization.

Table 2: Experimental Configuration Comparison

Parameter	Baseline	Enhanced
Enhanced Evaluation	False	True
Feedback Enabled	False	True
Target Threshold	0.5	0.5
Iterations	3	3
Memory Window	5	5

5.2 Performance Results

Our experimental analysis revealed significant efficiency improvements with the enhanced configuration while maintaining equivalent jailbreaking effectiveness:

Table 3: Experimental Performance Comparison

Metric	Baseline	Enhanced
Success Rate	100%	100%
Total Execution Time	13,874s	8,449s
Convergence Speed Improvement	—	+50.0%
Processing Speed Ratio	—	1.6x
Category-Specific Performance	—	-37.5%

5.3 Key Findings

The experimental results demonstrate several critical insights:

- **Maintained Effectiveness:** Both configurations achieved 100% success rates for jailbreaking the target model, indicating that enhanced features do not compromise adversarial capability
- **Significant Speed Improvement:** The enhanced configuration completed jailbreaking

tasks in 60% of the baseline time (8,449s vs 13,874s)

- **Processing Efficiency:** 1.6x improvement in processing speed ratio suggests more targeted prompt evolution
- **Category-Specific Behavior:** 37.5% reduction in homophobic category performance may indicate inherent safety biases in AI-based evaluation

5.4 Convergence Analysis

The enhanced configuration demonstrated superior convergence characteristics:

- More binary score patterns (clear success/failure vs. marginal scores)
- Faster identification of effective attack vectors
- Reduced trial-and-error iterations through intelligent feedback
- Consistent performance across different prompt categories

6 Discussion

6.1 Implications for Adversarial AI Research

Our results demonstrate that enhanced evolutionary approaches can maintain jailbreaking effectiveness while dramatically improving research efficiency. The 50% improvement in convergence speed suggests that AI-powered feedback systems can significantly accelerate vulnerability discovery without compromising success rates.

6.2 Efficiency vs. Effectiveness Trade-offs

The experimental data reveals important considerations for adversarial research:

- **Time Efficiency:** Enhanced features reduce jailbreaking discovery time by 40%, enabling more comprehensive security assessments
- **Maintained Success:** 100% success rates for both configurations confirm that optimization improvements do not reduce adversarial capability

- **Processing Intelligence:** 1.6x speed improvement indicates more targeted evolution rather than brute-force iteration
- **Category Sensitivity:** Reduced performance in homophobic categories may reflect inherent biases in AI evaluation systems

6.3 AI Evaluator Behavior Analysis

The 37.5% reduction in homophobic category performance reveals potential limitations:

- AI evaluators may have built-in safety mechanisms that resist certain harmful categories
- This could create blind spots in comprehensive vulnerability assessment
- Researchers should be aware of potential evaluation biases when using AI-powered systems
- Multiple evaluation approaches may be necessary for complete coverage

6.4 Practical Research Applications

These findings have direct implications for red-teaming and security research:

- **Accelerated Discovery:** Researchers can identify model vulnerabilities in significantly less time
- **Resource Optimization:** Improved efficiency allows for broader testing across more models and categories
- **Quality Assessment:** More decisive score patterns help distinguish successful from failed attacks
- **Systematic Evaluation:** Framework enables reproducible vulnerability assessment protocols

6.5 Limitations

Several limitations should be noted in this preliminary evaluation:

- Evaluation limited to English-language prompts
- Focus on text-based attacks excludes multimodal vulnerabilities
- Computational requirements may limit accessibility for some researchers

- Ethical considerations require careful deployment guidelines
- **Resource Constraints:** Due to computational and time limitations, experiments were restricted to 3 iterations with single-category testing
- **Limited Experimental Scope:** Current evaluation covers only a small subset of possible configurations and attack strategies
- **Short-Term Assessment:** Extended experiments over longer periods may reveal different performance patterns and convergence behaviors

7 Future Work

7.1 Extended Experimental Evaluation

Future research should address the current experimental limitations:

- **Longer Iteration Cycles:** Extended experiments with 10+ iterations to assess long-term convergence patterns and stability
- **Multi-Category Assessment:** Comprehensive evaluation across all toxicity categories simultaneously
- **Large-Scale Studies:** Systematic evaluation across multiple model architectures and sizes
- **Temporal Analysis:** Long-running experiments to understand performance evolution over extended periods

7.2 Technical Extensions and Improvements

Future development directions include:

- **Multimodal Support:** Extending to image and audio inputs
- **Cross-Lingual Evaluation:** Supporting multiple languages
- **Real-Time Adaptation:** Dynamic strategy adjustment during evaluation
- **Federated Learning:** Collaborative vulnerability discovery across institutions

7.3 Defense Development

The insights gained from EvoTox can inform defensive strategies:

- Adversarial training using discovered attack patterns
- Dynamic safety filtering based on evolutionary trends
- Proactive vulnerability scanning for deployed models

8 Conclusion

We have presented EvoTox, a comprehensive evolutionary framework for LLM safety evaluation that significantly advances the state-of-the-art in automated red-teaming. Our preliminary evaluation demonstrates substantial improvements in vulnerability discovery efficiency while maintaining jailbreaking effectiveness.

The controlled comparison between baseline and enhanced configurations reveals a 50% improvement in convergence speed with 100% success rates for both approaches. The combination of evolutionary optimization, enhanced evaluation with explanations, and memory-based learning creates a powerful platform for systematic safety assessment, achieving 1.6x processing speed improvements through intelligent feedback mechanisms.

While resource and time constraints limited our current evaluation to short-term experiments with restricted scope, the results provide strong evidence of EvoTox’s potential for advancing adversarial AI research. The framework’s efficiency gains suggest that extended experiments with longer iteration cycles and comprehensive multi-category assessments could yield even more significant insights into model vulnerabilities and defense strategies.

Most importantly, EvoTox provides researchers and practitioners with actionable insights for improving model safety through accelerated vulnerability discovery. The detailed explanations and pattern analysis enable targeted defense development, contributing to the broader goal of safe and beneficial AI systems.

Future work involving extended experimental campaigns with increased computational resources promises to unlock the full potential of this evolutionary approach, potentially revealing deeper insights into model behavior and more effective adversarial strategies.

Our open-source implementation¹ enables reproducible research and collaborative advancement of AI safety evaluation methodologies.

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References

- [1] OpenAI. GPT-4 Technical Report. arXiv preprint arXiv:2303.08774, 2023.
- [2] Andy Zou, Zifan Wang, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. arXiv preprint arXiv:2307.15043, 2023.
- [3] Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. Jailbroken: How does llm safety training fail? arXiv preprint arXiv:2307.02483, 2023.
- [4] Ethan Perez, Saffron Huang, Francis Song, Trevor Cai, Roman Ring, John Aslanides, Amelia Glaese, Nat McAleese, and Geoffrey Irving. Red teaming language models with language models. arXiv preprint arXiv:2202.03286, 2022.
- [5] Samuel Gehman, Suchin Gururangan, Maarten Sap, Yejin Choi, and Noah A. Smith. RealToxicityPrompts: Evaluating neural toxic degeneration in language models. In Findings of the Association for Computational Linguistics: EMNLP 2020, pages 3356-3369, 2020.
- [6] Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Constitutional ai: Harmlessness from ai feedback. arXiv preprint arXiv:2212.08073, 2022.
- [7] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. Advances in Neural Information Processing Systems, 35:27730-27744, 2022.

¹Available at: <https://github.com/Pecneb/EvoTox>

- [8] Deep Ganguli, Liane Lovitt, Jackson Kernion, Amanda Aspell, Yuntao Bai, Saurav Kadavath, Ben Mann, Ethan Perez, Nicholas Schiefer, Kamal Ndousse, et al. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. arXiv preprint arXiv:2209.07858, 2022.
- [9] Eric Wallace, Shi Feng, Nikhil Kandpal, Matt Gardner, and Sameer Singh. Universal adversarial triggers for attacking and analyzing nlp. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pages 2153-2162, 2019.